SET 9 – MATHEMATICS – APPLICATIONS OF SUMS
ANSWER KEY & MARKING SCHEME

Total Marks: 15 | Model Answers & Step-by-Step Breakdown

Q1. Sum of First 100 Natural Numbers (2 Marks)

MODEL ANSWER

$\Sigma n = n(n + 1)/2 = 100 \times 101 / 2 = 5050$

Answer: 5050

KEYWORD MARKING SCHEME

Marks	Awarded for
1.0	Correct formula.
1.0	Correct substitution and answer 5050.

Q2. Sum of Cubes up to 20 (2 Marks)

MODEL ANSWER

Find $1^3 + 2^3 + \ldots + 20^3$.

$= (20 \times 21 / 2)^2 = 210^2 = 44100$

Answer: 44100

KEYWORD MARKING SCHEME

Marks	Awarded for
1.0	Correct formula $[n(n + 1)/2]^2$.
1.0	Correct answer 44100.

Q3. Sum of Cubes in a Range (3 Marks)

MODEL ANSWER

Find $11^3 + 12^3 + \ldots + 20^3$.

$= (1^3 + \ldots + 20^3) - (1^3 + \ldots + 10^3)$

$= 44100 - 3025 = 41075$

Answer: 41075

KEYWORD MARKING SCHEME

Marks	Awarded for
1.0	Writing the difference of two sums.
1.0	Using 44100 and 3025 correctly.
1.0	Final answer 41075.

Q4. Finding n from the Sum (3 Marks)

MODEL ANSWER

If $1^3 + 2^3 + \ldots + n^3 = 2025$, find n.

$[n(n + 1)/2]^2 = 2025 \Rightarrow n(n + 1)/2 = 45$

$n(n + 1) = 90 \Rightarrow n^2 + n - 90 = 0 \Rightarrow (n - 9)(n + 10) = 0$

n = 9 (rejecting n = −10, since n is a natural number)

KEYWORD MARKING SCHEME

Marks	Awarded for
1.0	Equating $[n(n + 1)/2]^2 = 2025$.
1.0	Forming and solving the quadratic.
1.0	Correct n = 9, rejecting the negative root.

Q5. Sum of k(k + 1) (5 Marks)

MODEL ANSWER

Find a formula for 1×2 + 2×3 + 3×4 + ... + n(n + 1) and evaluate it for n = 10.

Split the general term:

$k(k + 1) = k^2 + k$

$Sum = \Sigma k^2 + \Sigma k = n(n + 1)(2n + 1)/6 + n(n + 1)/2$

$= n(n + 1)[(2n + 1) + 3] / 6 = n(n + 1)(2n + 4)/6 = n(n + 1)(n + 2)/3$

For n = 10:

10 × 11 × 12 / 3 = 440

KEYWORD MARKING SCHEME

Marks	Awarded for
1.0	Splitting $k(k + 1) = k^2 + k$.
1.0	Using the formulas for Σk^2 and Σk.
1.5	Correct simplification to n(n + 1)(n + 2)/3.
1.5	Correct evaluation for n = 10: 440.

FINAL STRUCTURE

Set	Subject	Questions	Marks
Set 7	Mathematics	5	15
Set 8	Mathematics	5	15
Set 9	Mathematics	5	15
Total	**—**	**15**	**45**